Last Time

Wilcoxon's Rank Tests

- Wilcoxon's Matched-Pairs Signed-Ranks Test

χ^2 Tests of Goodness of Fit and Association

- Goodness of Fit Example
- Test of Association Example

Today

- Testing Equality of Variances
  - The F-Distribution
- Introduction to ANOVA
- Logic of ANOVA

Contingency Table Example

Does IV drug use increase the rate of HIV infection in a population?

	HIV positive	HIV negative	Total
No IV Drug Use	4	73	77
IV Drug Use	15	33	48
Total	19	106	*125*

Recall the assumption of Homoscedasticity

For the independent sample t-test, assume

- Measured variable (X) is normally distributed
- Means (samples) $\overline{X_1}$ and $\overline{X_2}$ are independent
- Population variances are equal ($\sigma_1^2 = \sigma_2^2$)

How do we test for homoscedasticity?

Tests for Equality of Variances

- Levene's test (DescTools::LeveneTest(money$\sim$happy))
- Fligner-Killeen test (stats::fligner.test(money$\sim$happy)
- F ratio (stats::var.test(money$\sim$happy))
- Bartlett test (stats::bartlett.test(money$\sim$happy))

The Null Hypothesis

$$
\begin{aligned}
H_0 :& \quad \sigma_1^2 = \sigma_2^2 \\
H_1 :& \quad \sigma_1^2 \neq \sigma_2^2
\end{aligned}
$$

The Null Hypothesis

$$H_0: \quad \sigma_1^2 = \sigma_2^2$$
$$H_1: \quad \sigma_1^2 \neq \sigma_2^2$$

Example: Sikes and Westefeld (2002)
Does therapist posture influence a client's rating of the therapist's expertise?

Group 1 Straight	Group 2 Slouched
$\overline{X}_1 = 59.23$	$\overline{X}_2 = 57.35$
$s_1 = 12.37$	$s_2 = 14.67$
$N_1 = 48$	$N_2 = 52$

The Null Hypothesis Reformulated

Group 1	Group 2
$s_1 = 12.37$	$s_2 = 14.67$
$N_1 = 48$	$N_2 = 52$

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Note that $H_0: \ \sigma_1^2 = \sigma_2^2$ is equivalent to $H_0: \ \frac{\sigma_1^2}{\sigma_2^2} = 1$.

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Note that $H_0: \ \sigma_1^2 = \sigma_2^2$ is equivalent to $H_0: \ \frac{\sigma_1^2}{\sigma_2^2} = 1$.

We know that s_j^2 is a "good" estimate of σ_j^2.

Therefore we will test H_0 using the statistic $\frac{s_1^2}{s_2^2}$.

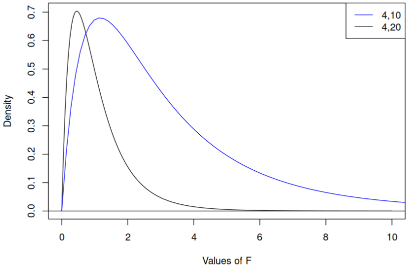
The F-ratio or F-statistic

$$F = \frac{s_1^2}{s_2^2}$$

is distributed as an F with $N_1 - 1$ (numerator) and $N_2 - 1$ (denominator) degrees of freedom if H_0 is true.

Characteristics of the F Distribution

- F is always positive
- Positively skewed
- Two parameters: numerator and denominator degrees of freedom



One-tailed Test

Test $H_0 : \sigma_1^2 = \sigma_2^2$ versus $H_1 : \sigma_2^2 > \sigma_1^2$

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$s_1 = 12.37$	$s_2 = 14.67$
$N_1 = 48$	$N_2 = 52$

$$
\begin{aligned}
F &= \frac{s_2^2}{s_1^2} \\
&= \frac{14.67^2}{12.37^2}
\end{aligned}
$$

One-tailed Test

Test $H_0: \sigma_1^2 = \sigma_2^2$ versus $H_1: \sigma_2^2 > \sigma_1^2$

Group 1	Group 2
$s_1 = 12.37$	$s_2 = 14.67$
$N_1 = 48$	$N_2 = 52$

$$F = \frac{s_2^2}{s_1^2} = \frac{14.67^2}{12.37^2} = \frac{215.21}{153.02} = 1.41$$

Critical F Value vs. P-value

If H_0 is true, s_2^2/s_1^2 is distributed as an F with

numerator degrees of freedom	$52 - 1 = 51$
denominator degrees of freedom	$48 - 1 = 47$

Critical F Value vs. P-value

If H_0 is true, s_2^2/s_1^2 is distributed as an F with

numerator degrees of freedom $52 - 1 = 51$

denominator degrees of freedom $48 - 1 = 47$

```
> qf(0.95,51,47)

[1] 1.612235
```

$F = 1.41 \rightarrow$ Fail to reject H_0.

```
> 1 - pf(14.67^2/12.37^2,51,47)

[1] 0.1195602
```

$0.12 > \alpha \rightarrow$ Fail to reject H_0.

GSS Attitude Question

```
> Income as a function of Attitude
> X <- data$REALRINC[data$YOUNGEN==1]
> Y <- data$REALRINC[data$YOUNGEN==2]
> var.test(X,Y)


 F test to compare two variances

data:  X and Y
F = 0.79146, num df = 336, denom df = 535, p-value = 0.0191
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.6538117 0.9622324
sample estimates:
ratio of variances
         0.7914581
```

Two-tailed Test

- $F = \dfrac{\text{Larger } s^2}{\text{Smaller } s^2}$
- Don't forget to get the numerator and denominator dfs right.
- Cut α in half: If p-value < 0.025 then reject for an $\alpha = 0.05$ level test.

(You don't have to worry about this too much in R, but F-tables only give the upper tail values.)

F-table

APPENDIX F: CRITICAL VALUES OF THE F DISTRIBUTION

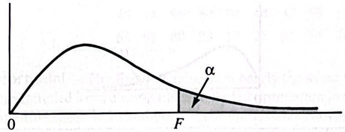

TABLE 1 $\alpha = .05$

Denominator	Degrees of Freedom for Numerator															
	1	2	3	4	5	6	7	8	9	10	15	20	25	30	40	50
1	161.4	199.5	215.8	224.8	230.0	233.8	236.5	238.6	240.1	242.1	245.2	248.4	248.9	250.5	250.8	252.6
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.43	19.44	19.46	19.47	19.48	19.48
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.70	8.66	8.63	8.62	8.59	8.58
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.86	5.80	5.77	5.75	5.72	5.70
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.62	4.56	4.52	4.50	4.46	4.44
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	3.94	3.87	3.83	3.81	3.77	3.75
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.51	3.44	3.40	3.38	3.34	3.32
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.22	3.15	3.11	3.08	3.04	3.02
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.01	2.94	2.89	2.86	2.83	2.80
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.85	2.77	2.73	2.70	2.66	2.64
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.72	2.65	2.60	2.57	2.53	2.51
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.62	2.54	2.50	2.47	2.43	2.40
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.53	2.46	2.41	2.38	2.34	2.31
14	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Introductory Remarks about ANOVA

- How do we compare several (more than two) populations?

$$H_0: \ \mu_1 = \mu_2 = \cdots = \mu_k$$

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$$H_0: \ \mu_1 = \mu_2 = \cdots = \mu_k$$

- Consider pairwise t-tests, i.e.

$$\begin{aligned} H_0 : &\quad \mu_1 = \mu_2 \\ H_0 : &\quad \mu_1 = \mu_3 \\ &\vdots \\ H_0 : &\quad \mu_{k-1} = \mu_k \end{aligned}$$

Problems

- For k populations,
$$\binom{k}{2} = k(k-1)/2$$
tests would be required.
  - E.g., if $k = 5$, 10 tests would be necessary.
- This is time consuming and tedious!

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- For k populations,
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tests would be required.
  - E.g., if $k = 5$, 10 tests would be necessary.
- This is time consuming and tedious!
- Consider the probability of making a Type I error.

Type I Error Rate

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\begin{aligned}
& p(\text{At least 1 Type I error}) \\
= \quad & 1 - p(\text{No Type I errors})
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$$

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\end{aligned}
$$

where m is the number of tests to be performed.

Type I Error Rate

$$\begin{aligned} & p(\text{At least 1 Type I error}) \\ & \quad = \quad 1 - p(\text{No Type I errors}) \\ & \quad = \quad 1 - (1-\alpha)^m, \end{aligned}$$

where m is the number of tests to be performed.
If each test has $\alpha = .05$, and $k = 5$ $(m = 10)$, then

$$1 - (1 - .05)^{10} = .4013!$$

Notes

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

- Each mean μ_i refers to the population mean for a particular treatment condition.
- Treatment conditions are called *levels* of an independent variable or *factor*.

An example using a *nominal* independent variable

- Patients with a fear of heights are randomly assigned to one of five psychotherapy treatments:
  - Gestalt
  - Psychoanalysis
  - Rational Emotive
  - Behavior Modification
  - No Treatment (Control)
- Measure ratings of fear after treatment.

An example using a *numerical* independent variable

- Rats learning a maze are given one of five levels of a cognitive-enhancing drug:
- 0 mg, 5 mg, 10 mg, 20 mg, 50 mg
- Measure number of cells in hippocampus.

Back to the Null Hypothesis

$$H_0: \; \mu_1 = \mu_2 = \cdots = \mu_k$$

- This null hypothesis is sometimes called the *omnibus* hypothesis.
- The ANOVA performed to reject H_0 is sometimes called the *omnibus test* or *omnibus F*.

Back to the Null Hypothesis

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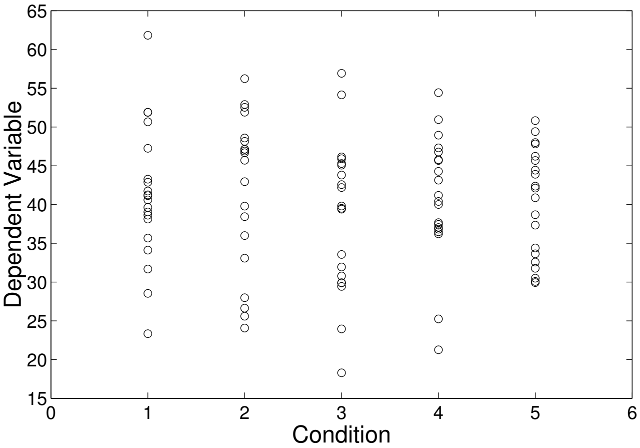
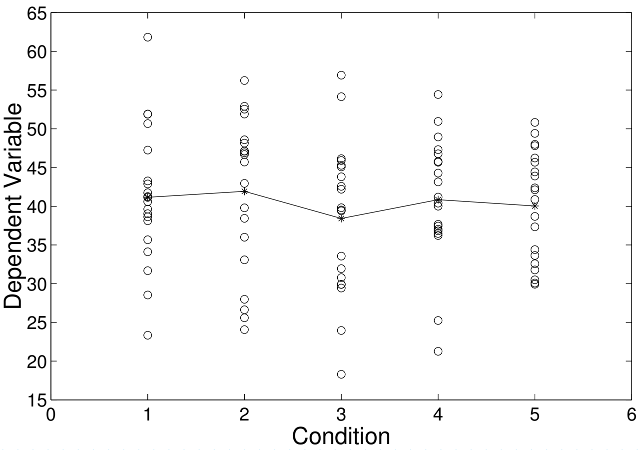
- This null hypothesis is sometimes called the *omnibus* hypothesis.
- The ANOVA performed to reject H_0 is sometimes called the *omnibus test* or *omnibus F*.
- The alternative hypothesis is $H_A : $ not H_0: at least one of the means is different from the others.

The logic of ANOVA

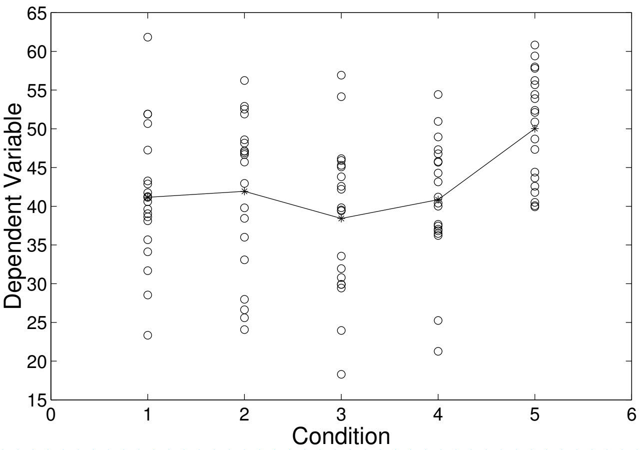
If all means are the same, the mean of each group should be about the same:



All population means equal



One population mean unequal



Consider the means alone

	1	2	3	4	5	Mean
H_0 true	41.16	41.92	38.43	40.85	40.03	40.48
H_1 false	41.16	41.92	38.43	40.85	50.03	42.48

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$$\begin{aligned} \overline{\overline{X_0}} &= 40.48 \\ \overline{\overline{X_1}} &= 42.48 \end{aligned}$$

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	1	2	3	4	5	Mean
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$$\begin{aligned}
\overline{\overline{X_0}} &= 40.48 \\
\overline{\overline{X_1}} &= 42.48 \\
s^2_{\overline{X_0}} &= 1.77 \\
s^2_{\overline{X_1}} &= 19.52
\end{aligned}$$

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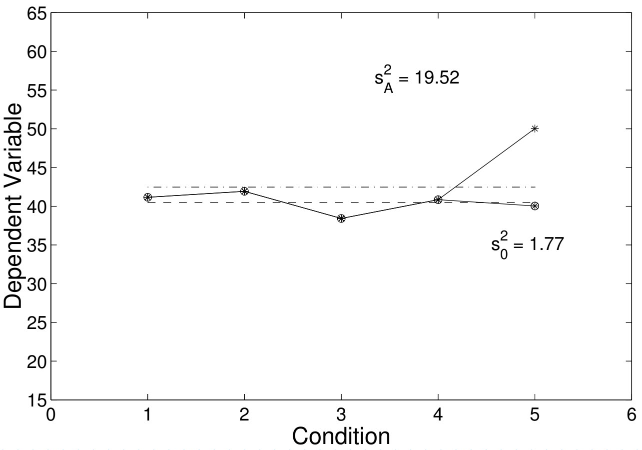
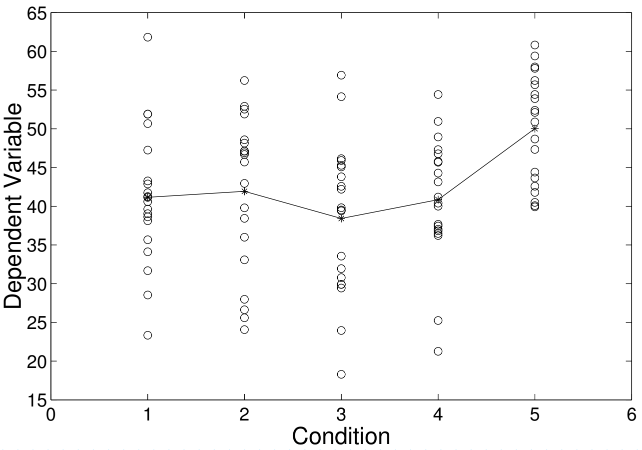
$$\begin{aligned}\overline{\overline{X_0}} &= 40.48 \\ \overline{\overline{X_1}} &= 42.48 \\ s^2_{\overline{X_0}} &= 1.77 \\ s^2_{\overline{X_1}} &= 19.52\end{aligned}$$

If H_0 is false, that fact will be reflected not only in differences between the sample means, but also in the *variance* of the sample means across conditions.

With pictures



Within-groups vs. Between-groups variance



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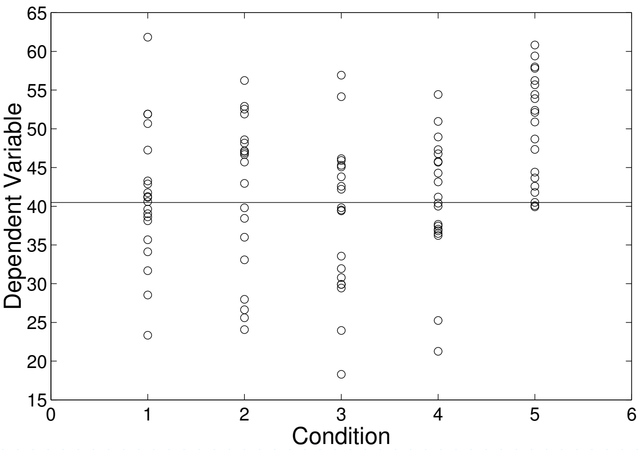
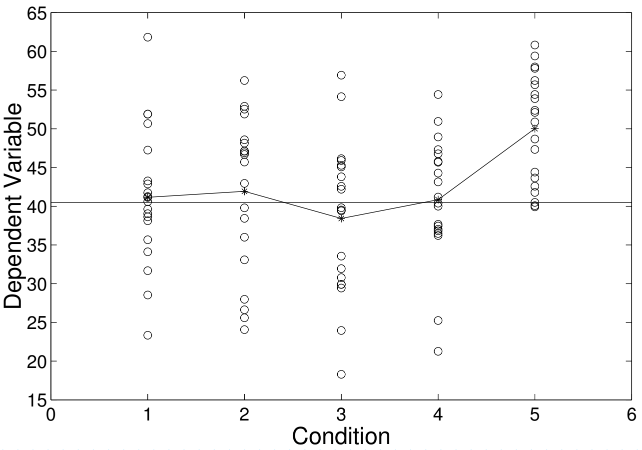


Partitioning the variance



Sources of variation

- Within-groups variance reflects the deviation of any score X_{ij} from its group mean $\overline{X_j}$.
- Between-groups variance reflects the deviation of any mean $\overline{X_j}$ from the overall mean $\overline{\overline{X}}$.

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- Between-groups variance reflects the deviation of any mean $\overline{X}_j$ from the overall mean $\overline{\overline{X}}$.
- Note to remember for later: Remember how s^2 is an estimate of σ^2? Within- and Between-groups variance are also estimates of population variance.